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Design of Accurate Multi-class Optimized Lightweight Convolution Neural Network for Rice Varieties Classification

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Abstract

Worldwide, more than 40k rice varieties are existing, each with different nutritional content and quality. Identifying these has to be consistent, automated, and accurate. Considering the feature extraction process, convolution Neural Networks (CNN) are preferred over machine learning (ML) for this classification. Transfer learning approaches help to optimize the CNN model; therefore, it fits in an FPGA. Seven different CNN models were proposed to classify five rice varieties, each model differs based on the: kernel depth; the number of convolution layers (CL); the number of fully connected layers (FCL); and the number of neurons per FCL. These were analyzed considering 70% and 30% for training and testing respectively. A dataset of 15,000 images/variety with each image of resolution. This results in an **O**ptimized **L**ight**w**eight **C**onvolutional **N**eural Network (OpLW-CNN) model, having a CL, followed by two FCLs. This model is further analyzed using a random set of images: 500, 5000, and 75000 to fit the model optimally. This model achieves 99%, 98.13%, and 98.14% of specificity, F1-score, and accuracy for a set of 5000 images. These metrics are approximately 1% to 2% lesser than the performance of the benchmark model, and 81.5% fewer computations. Also, this model requires less than a second to classify an image.

I. Introduction

Artificial Intelligence (AI) based multi-classification techniques have a wide range of applications such as the agricultural, medical, and industrial sectors. Computer vision-based tasks including classification, object detection, segmentation, and recognition have led to successful adaptation in the above applications. Particularly, in agriculture fields, the classification of the rice grains variety is an important factor in determining the quality of grains, acceptance of the market value rate, stability of the storage, and nutritious content. Based on these factors, rice varieties are priced on the market [1, 2, 3].

Manual evaluation or classification of grains can be time-consuming, expensive, and inconsistent due to limited evaluation assessment experts. Also, when an evaluation is conducted on a large scale, quick decision-making through manual methods may be challenging [4, 5]. Rice is one of the major grain products and it is consumed all over the world. After acquiring the digital images of the rice products, the classification of rice grains is characteristics based on various parameters such as texture, shape, and color. Several image processing algorithms are adopted to establish these characteristics and carry out classification operations.

In recent years, Machine learning (ML) and Deep Learning (DL) approaches have been effective in the accurate and quick evaluation of a massive volume of data. Using these techniques an automated, cost-effective, efficient, and non-destructive classification of rice varieties is being proposed [10–14]. This increases product quality and adheres to food safety standards [6].

A general block diagram of an image-based multi-classifier system is illustrated in Fig. 1. Image pre-processing and feature extraction techniques can be used to supply only the key features of the input images. These speed up the classification process accurately. Furthermore, it has been illustrated that these features can be used with ML-based classifiers, whereas DL-based classifiers do not use them.

In the past decade, some of the reported rice grain classifiers are ANN (Artificial Neural Network), SVM (Support Vector Machine), LR (Linear Regression), DT (Decision Tree), DNN (Deep Neural Network), and CNN (Convolution Neural Network) [7]. CNN models can automatically enhance and extract the features resulting in a top-level accuracy of near to 100% in all classification metrics. But large-scale data management and high-speed computing resources might be the significant challenges to implementing the CNN models into hardware such as Android phones, and embedded boards.

Based on the above challenges, this study aims to develop a precise and customized CNN model through a transfer learning approach. This will increase hardware performance as the computations are reduced considerably for rice varieties classification as follows:

1. Proposed 7 different CNN modes (Type 1 to Type 7) based on the transfer learning approach for multiclass rice grain classifier. Each model, comprises a different depth of filter weights, the number of the CL & FC layers, and the number of neurons per FC. Analysis of these models results from (a) the total number of computations required and (b) the classification metrics of the model. The best CNN model is chosen based on the trade-off between (a) and (b).
2. The **O**ptimised **L**ight**w**eight **C**onvolution **N**eural **N**etwork (OpLW-CNN) model is proposed with the structure of one CL layer along 8 filter depths and two FC layers along 100 & 5 neurons $(CL_1\,(8) : FC_1\,(100) - FC_2\,(5))$ using MATLAB-based environment. The OpLW- CNN has adopted the 25.5M computation and above 98% of overall performance metrics compared to benchmark CNN models.

The paper is organized as follows. The dataset, detailed expression of multiclass performance evaluations, and proposed lightweight CNN model for rice classification techniques are presented in Section II. The experimental findings from the analysis are discussed in Section II.

Finally, the conclusion and future works are in Section IV.

II. Materials and Methods

A conventional ML system for the classification of rice grain varieties should have the following operations: image resizing and acquisition, pre-processing, segmentation, extracting the number of features, and classification. The employment of CNN-based models has eliminated the additional feature extraction and segmentation techniques. Therefore, the CNN-based rice classifier model does the automatic extraction of unique features such as edges, texture, and color using the learnable parameters, and a SoftMax function helps to provide multi-classification accurately.

A. Dataset Description and Experimental Setup

The study analyses the data sets from five commonly grown rice varieties in Turkey: Arborio, Basmati, Ipsala, Jasmine, and Karacadag. Each one of these is shown in Fig. 2. The dataset contains a total of 75,000 images of rice grains, with 15,000 images for each rice variety.

Each image has a resolution of 250 × 250 × 3 pixels [13]. The CNN models use RGB images for training and classification. During the training of our model, the training dataset is 70%, and the remaining 30% for testing the model. The experiments were performed using an Intel (R) Processor clocked at 3.20 GHz, and 16 GB of RAM with MATLAB R2020b environment running on a 64-bit Linux operating system.

B. Performance Evaluation for Multiclassification Model

The classification metrics of a multiclassification model are statistically calculated from a unique confusion matrix. The confusion matrix records the number of test data outcomes as the correct and incorrect classification of positive samples as well as the correct and incorrect categorization of negative samples [12]. In this paper, the dataset and proposed model has adopted for five classes of rice varieties therefore the number of test data outcomes is five. A structure of the five classes confusion matrix with labels A, B, I, J & K (The first letter of each rice variety) has been illustrated in Fig. 3. Also, in Fig. 3, the expression of each class statistical calculation (TP, FP, TN, & FN) for multi-class analysis has been formulated. $O_{i,j}$ represents the number of sample data placed on the position of the matrix index after training or testing the model. i and j denote the actual and predicted classes which vary from 1 to 5. Also, y varies similarly as the classifier classifies 5 varieties of rice.

Table I displays the metrics derived through statistical calculations for the five-class confusion matrix, the formulae employed to derive these metrics, and details on the usage of the metrics. The goal of this paper is to identify the CNN model that is adapted to classify rice grains in a highly accurate, quick, and cost-effective manner. Since this is a multi-class classification problem, macro-averaged accuracy, precision, recall, and F1-score are the metrics used to compare the performance of various models as displayed in Table I. The macro-average metrics are the sum of all individual metrics computed across all the *N* classes [3, 9].

Table I Macro average performance metrics for *N*-classification

Metrics	Expression	Macro Average metrics	Description
Accuracy (A)	$\frac{TP(y)+TN(y)}{TP(y)+TN(y)+FP(y)+FN(y)}$	$\frac{1}{N}\sum_{y=1}^{N} A(y)$	“A” reveals the model's classification success.
Specificity (S)	$\frac{TN(y)}{TN(y)+FP(y)}$	$\frac{1}{N}\sum_{y=1}^{N} S(y)$	“S” provides the number of negative predictions that are accurately classified
Recall (R)	$\frac{TP(y)}{TP(y)+FN(y)}$	$\frac{1}{N}\sum_{y=1}^{N} R(y)$	“R” provides the number of positive predictions that are accurately classified
F1-score (F)	$\frac{2\times P(y)\times R(y)}{P(y)+R(y)}$	$\frac{1}{N}\sum_{y=1}^{N} F(y)$	"F" gives the values for Precision and recall harmonic mean. it provides feedback on whether the model is inaccurate for uneven distribution datasets.
Precision (P)	$\frac{TP(y)}{TP(y)+FP(y)}$	$\frac{1}{N}\sum_{y=1}^{N} P(y)$	“P” provides a positive estimation value.

C. Analysis of the CNN Models

a) Automatic features extraction

The CNN algorithm can perform as an end-to-end classifier. The CNN network can extract layers of features from the input images. Later using the extracted features, it may learn and classify the test input. Most of the references [11, 12] have used the trained CNN network structure such as AlexNet, VGG-16 & 19 to classify the variety of rice. The references [9, 13] adopted the transfer learning approach on these standard models and modified them at the final FC layer into 5 neurons based on the number of output classifications. These works have a dense model with many computations, learning parameters, and filter depth as same as in benchmark CNN models.

Here, a CNN model comprising one or more feature extraction layers (convolution layer - activation layer - pooling layer), followed by one or more fully connected layers, and finally a classification layer (SoftMax layer) [9, 13] is considered for the analysis. The convolution layer extracts maximum features based on the depth of the filters, the activation layer fixes the range of these features by thresholding, and the pooling layer minimizes the features based on a strategy (max-pooling, Average-pooling) and provides input for the following layers. Having similar feature extraction layers should help to improve the classification metrics further. To converge the features into the required classes, two fully connected layers were used. The first FC layer converges the extracted features to 1000/100 features. The second FC layer further converges these features into five output features. Finally, the SoftMax layer statistically interprets the features into the output classes. The investigated structure (St) of the CNN is as follows:

$$St = (Fd) - CL_2\,(Fd)\ldots CL_n\,(Fd) : FC_1\,(Nr) - FC_2\,(Nr)\ldots FC_n\,(Nr)$$

1

where$FC_n \& CL_n$ refers to the number of convolutional layers and fully connected layers, Fd filter depth. All the kernels were 3 × 3 in size. Here, every convolutional layer was followed by ReLU and the max polling layer. Nr refers to the number of neurons in FC layers. The number of neurons in the last FC layer has fixed to five Nr due to the number of our classes.

Considering these, seven different CNN models were proposed from type 1 to type 7 as shown in Table II. These models were analyzed for the computational complexity and resulting classification metric which is provided in Table III.

b) Trade-off analysis of different CNN models

These seven different structures of the CNN model for rice classification have been analyzed based on the classification metrics: accuracy, F1-score, recall, precision, and specificity. Also, each model has been analyzed for its computational complexity (number of MAC operations). Using the Stochastic Gradient Descent with Momentum (SDGM) Optimizer, the models were trained using categorical cross-entropy. The learning rate is set to 0.0001. Except for the learning rate, the optimizer's parameters were fixed to the default values as

mentioned in the standard CNN models. A total of 30 epochs are allowed for each model to train. With the patience of 15 Epochs, early stopping is also applied while monitoring the validation score. The best epoch for each model was decided with the lowest validation loss. The appropriate case of the model is chosen as the best instance for the final assessment of the testing phase.

The computation complexity for the CNN models is calculated based on the kernel size (KS), number of operations required per kernel with input (R), size of the input (I), and number of depths (D) required.

$$Computationperlayer = I \times KS \times D \times R$$

2

For example, In the Type 1 model, the CL_1 contains the input size of I = 250×250×3, kernel size with a depth of KS = 3×3, its depth D = 8, R = 9 operations required per kernel size with input. It produces 13.28×10^6 operations per CL layer. Likewise, the other layer's computations in type 1 are estimated and accumulated together. Similarly, Type 2 to Type 7 computation complexity has been estimated and noted in Table II.

Table II Computation complexity analysis for the types of the CNN model.

Model	CL_1	CL_2	CL_3	FC_1	FC_2	Total Computation ($M = 10^6$)
	Depth, #Computation			Neurons, #Computation		
Type 1	8, 13.28M	16, 17.14M	32, 16.04M	100, 2.88M	5, 500	49.3M
Type 2	8, 13.28M	16, 17.14M	-	100, 5.95M	5, 500	36.3M
Type 3	8, 13.28M	16, 17.14M	-	1000, 59.53M	5, 5000	89M
Type 4	16, 26.56M	32, 68.58M		100, 11.90M	5, 500	107M
Type 5	8, 13.28M	32, 34.85M	-	100, 11.90M	5, 500	60M
Type 6	4, 6.64M	-	-	1000, 61.50M	5, 5000	68M
Type 7	**8, 13.28M**	**-**	**-**	**100, 12.3M**	**5, 500**	**25.5M**

Note: Type 1: $\mathrm{CL}_1\,(8) - \mathrm{CL}_2\,(16) - \mathrm{CL}_3\,(32) : \mathrm{FC}_1\,(100) - \mathrm{FC}_2\,(5)$; **Type 2**: $\mathrm{CL}_1\,(8) - \mathrm{CL}_2\,(16) : \mathrm{FC}_1\,(100) - \mathrm{FC}_2\,(5)$; **Type 3**: $\mathrm{CL}_1\,(8) - \mathrm{CL}_2\,(16) : \mathrm{FC}_1\,(1000) - \mathrm{FC}_2\,(5)$; **Type 4**: $\mathrm{CL}_1\,(16) - \mathrm{CL}_2\,(32) : \mathrm{FC}_1\,(100) - \mathrm{FC}_1\,(5)$; **Type 5**: $\mathrm{CL}_1\,(8) - \mathrm{CL}_2\,(32) : \mathrm{FC}_1\,(100) - \mathrm{FC}_2\,(5)$; **Type 6**: $\mathrm{CL}_1\,(4) : \mathrm{FC}_1\,(1000) - \mathrm{FC}_2\,(5)$; **Type 7**: $\mathrm{CL}_1\,(8) : \mathrm{FC}_1\,(100) - \mathrm{FC}_2\,(5)$.

Table III Classification metrics of the different types of the CNN model.

Model Type	Accuracy	F1-Score	Specificity	Recall	Precision
1	0.9932	0.9900	0.9983	0.9930	0.9900
2	0.9833	0.9833	0.9958	0.9833	0.9834
3	0.9800	0.9819	0.9955	0.9800	0.9810
4	0.9953	0.995	0.9988	0.9953	0.9900
5	0.9920	0.9922	0.9981	0.9920	0.9923
6	0.9768	0.9769	0.9942	0.9768	0.9771
7	**0.9864**	**0.9868**	**0.9967**	**0.9800**	**0.9800**

During testing, the models used 5000 input images per rice variety which is 30% of the total image data set. Each trained model has undergone testing to classify the test images and estimated their metrics. The confusion matrix of each multiclass model is derived and it

is used for the statistical calculation of TP, TN, FP, and FN as discussed in above Section B. The classification metrics of all the proposed types 1 to 7 are presented in Table III.

Analyzing the classification metrics presented in Table III, the models of type 1 to 7 have slightly dropped in the accuracy, Recall, and F1-Score in the range from 0.5–2.3% compared to the benchmark standard CNN model [12]. Likewise, Precision and Specificity have reduced from 0.2–1.3%. The best model based on the classification metrics is type 4. From the computation complexities presented in Table II, the type 1 to 7 models could require a number of computations ranging from 107×10^6 to 25.5×10^6. Considering all the proposed models, Type 7 is considered to be better in terms of computation complexity and classification metrics.

Therefore, the CNN structure of type 7 $(CL_1\,(8) : FC_1\,(100) - FC_2\,(5))$ should be the best model to implement in the hardware platforms such as FPGA. This model is named as “Optimised Lightweight CNN (OpLW-CNN) model”. The OpLW-CNN is implemented with (1) Feature Learning & (2) Classification stage as shown in Fig. 4. Stage 1 learning the features automatically using the filter/learnable parameters. It has one CL layer with a depth of 8 filter weights, followed by the Rectified Linear Unit (RELU) and pooling layer. Stage 2 classifies the features using two fully connected layers with 100 neurons first and 5 neurons later. It also has the SoftMax layer used to predict the order of the class for the rice variety. The computation complexity, parameters, and layers of the proposed OpLW-CNN are noted in Table IV.

Table IV Layers, parameters, and computation complexity of Type 7 model.

Layer type	Kernel Size	Learnable parameters	Stride	Output channel	Output size	Activation function	Number of computations
2D-Convolution	3×3	224	1	8	248×248	RELU	13.26M
Max Pooling	2×2	-	2	8	124×124	-	-
Fully Connected	1×1	12300900	-	100	100	RELU	12.3M
	1×1	505	-	5	5	SoftMax	
Total		**12.3M**					**25.56M**

III. Results and Discussion

In this section, the proposed OpLW-CNN model is compared with the existing models. The dataset consists of 15,000 images of each rice variety such as Arborio, Basmati, Ipsala, Jasmine, and Karacadag. The training and testing phases use 70% and 30% of the corresponding images respectively. The standard CNN model computation flow has been utilized to train the proposed CNN model using the transfer learning technique to increase the success rate and decrease the computation units. The OpLW-CNN approach has a classification success rate of 98.64%. Table V and Table VI display the confusion matrix for training and testing the OpLW-CNN model. Using this confusion matrix, statical values of TP, FP, TN, & FN are calculated for the classification metrics evaluation as shown in Fig. 3. The predicted labels of rice varieties classification are displayed in Fig. 5.

The OpLW-CNN model with the same computational complexity is further trained with 5000 input images. This is to verify the model’s training accuracy with fewer data. The classification metrics were reduced further but the reduction was within 2%. This is repeated with 500 sample images and it reduced the metrics to a minimum of 2% and a maximum of 5%. Therefore, the model can classify the rice varieties even with lesser input images. The results for these are presented in Table VII.

Table V OpLW-CNN model confusion matrix during training.

OpLW-CNN Model		Predicted Class				
		Arborio	Basmati	Ipsala	Jasmine	Karacadag
Actual Class	Arborio	9767	0	10	19	204
	Basmati	1	9847	0	152	0
	Ipsala	13	0	9950	37	0
	Jasmine	44	312	14	9629	1
	Karacadag	104	0	0	0	9876

Table VI OpLW-CNN model confusion matrix during testing.

OpLW-CNN Model		Predicted Class				
		Arborio	Basmati	Ipsala	Jasmine	Karacadag
Actual Class	Arborio	4968	0	3	15	14
	Basmati	0	4916	0	84	0
	Ipsala	7	0	4990	3	0
	Jasmine	14	74	15	4897	0
	Karacadag	102	0	0	0	4898

The proposed OpLW-CNN model has been compared with the existing models as in Table VII. The benchmark is the standard VGG-16 CNN model presented in [12]. Another transfer learning-based lightweight CNN model is proposed in [11]. It uses AlexNet as the benchmark and the size of the input images as 200×200. Also, the filter depth varies from 96 to 384 with kernel size from 11×11 to 3×3 and the number of layers is the same as the standard AlexNet model. The computation complexity of the [11] is calculated as explained in Eq. 2 and it is found to be 61M. This model is compared in terms of classification metrics with other ML models.

Table VII Studies with rice varieties classification in literature and comparison of this study.

Existing works	**Dataset (No. of images)**	**Output classes**	**Classifier**	**Accuracy (%)**	**F1-Score**	**Specificity**	**Recall**	**Number of Computation**
[12]	75000	5	CNN	100.00	1.000	1.000	1.000	138M
[10]	200	3	CNN	88.07	-	-	-	144M
[13]	3810	2	LR	93.02	0.9180	0.9358	0.9226	-
[11]	7399	3	DCNN	95.50	0.9525	0.9807	0.9735	61M
[14]	843	16	SVM	87.18	-	-	-	-
Proposed	75000	5	OpLW-CNN $CL_1(8) : FC_1(100) - FC_2(5)$	98.64	0.9868	0.9967	0.9800	25.5M
Proposed	5000	5	OpLW-CNN $CL_1(8) : FC_1(100) - FC_2(5)$	98.14	0.9813	0.9953	0.9814	25.5M
Proposed	500	5	OpLW-CNN $CL_1(8) : FC_1(100) - FC_2(5)$	95.64	0.9564	0.9891	0.9560	25.5M
'-' Unavailability of data								

Compared to [11], the proposed model accepts a much larger image of 250×250 and it is considering the VGG-16 as the benchmark model. The proposed OpLW-CNN model outperforms the 2.6% in accuracy & F1-Score; 2.3 times less computation compared to the DCNN model [11]. [13, 14] uses the linear regression and support vector machine approach for classification, and feature extraction is done by an additional image processing algorithm. Therefore, the proposed model is outperforming the classification metrics by 6% and 13% in terms of accuracy. [10] adopted the three CNN models such as VGG-16, VGG-19, and ResNet50. From these models, [10] claims that VGG-19 is considered to be better with an accuracy of 88%. The computation of the VGG-19 is estimated and attains to be 144M. It is 5.6 times greater than the proposed OpLW-CNN model. The overall classification metrics of the proposed OpLW-CNN model achieve more accuracy of 2.6–13.4% [10, 11, 13, 14] and reduces the computation resources of 58–82% compared to benchmark CNN models [10–14].

IV. Conclusion

This paper presents **O**ptimised **L**ight**w**eight-**C**onvolution **N**eural **N**etwork (**OpLW-CNN**) for the automatic multiclass rice varieties classification. To find the best structure, different transfer learning CNN models such as types 1 to 7 are developed and analyzed for classification metrics and the computational complexity (number of MAC operations). Comparing each model (type 1 to type 7), a model (Type 7) has maintained the classification metrics as closely as in the benchmark CNN model and requires 5 times lesser computational complexity. This model consists of one convolution layer, followed by two fully connected layers ($CL_1\left(8\right): FC_1\left(100\right) - FC_2\left(5\right)$) and denoted as OpLW-CNN. Also, the proposed OpLW-CNN model attains 0.9868 for F1-Score, 0.9967 for Specificity, 0.98 for Recall, and Precision. The average classification metrics of the OpLW-CNN have improved from 3–13% of Accuracy and 2.39 times to 5.64 times in computation operation compared to the existing works. Furthermore, the proposed OpLW-CNN can be implemented in the FPGA hardware with minimum memory usage and smaller computational complexity and at the same time meet the requirements of accurate classification in real-time.

Declarations

Authors Contribution

Deepika Selvaraj: Dataset collection; Design implementation; Statistical parameter analysis; Results comparison; Writing-original manuscript. Arunachalam Venkatesan: Work planning; Supervision; Validated the parameters; Revised the whole manuscript.

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Data Availability

The rice varieties dataset supporting the findings of this study is available within the article. Codes of the Op-LW CNN model during training and testing are available upon request from the corresponding author.

Conflict of Interest

The authors declare no competing interests.

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Figures

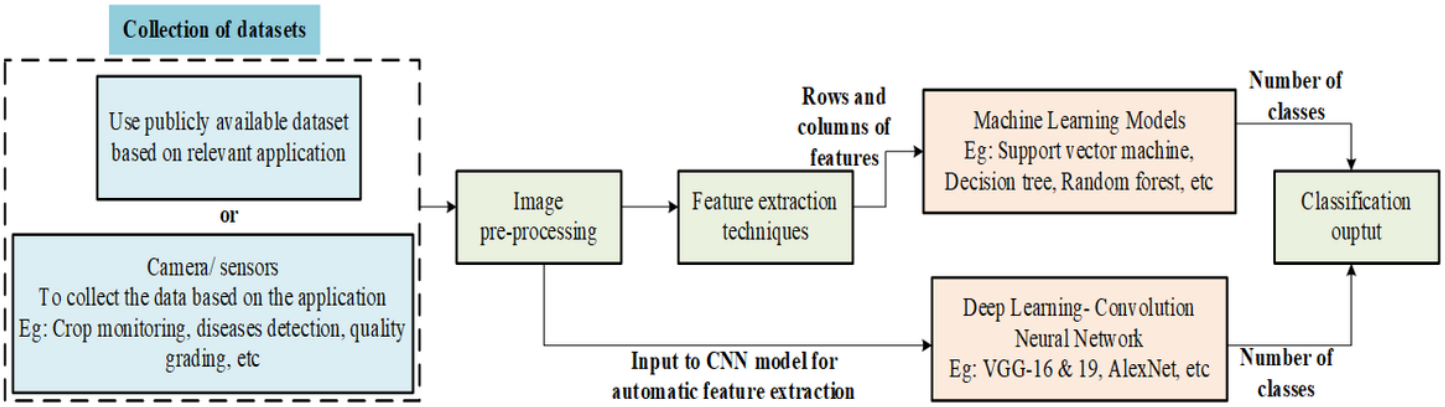

Figure 1

Methods followed in Existing works using ML and DL algorithms.

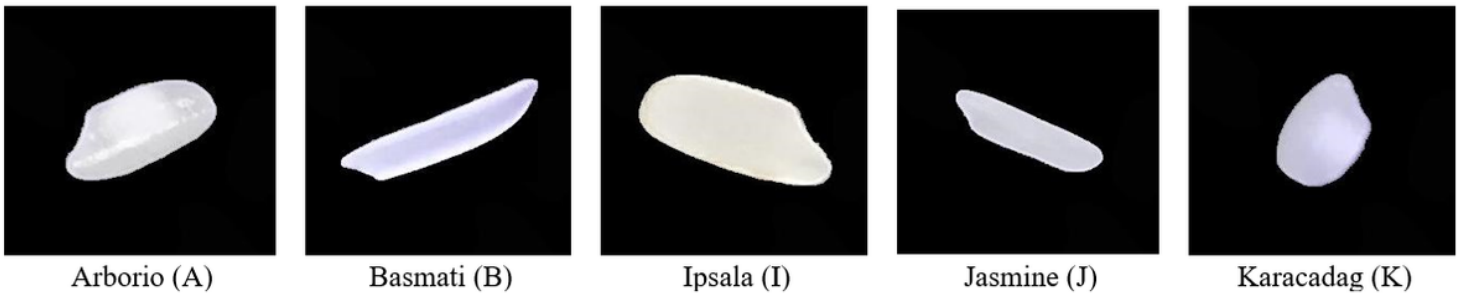

Figure 2

Variety of rice used in this experimental study.

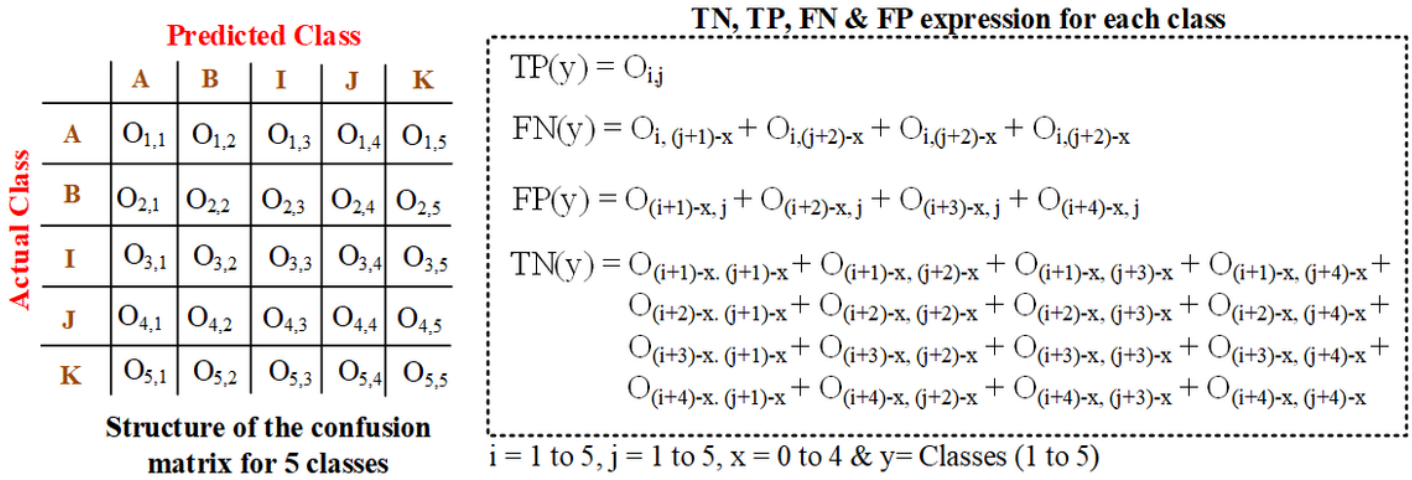

Figure 3

Multiclass confusion matrix & its expression for performance evaluation.

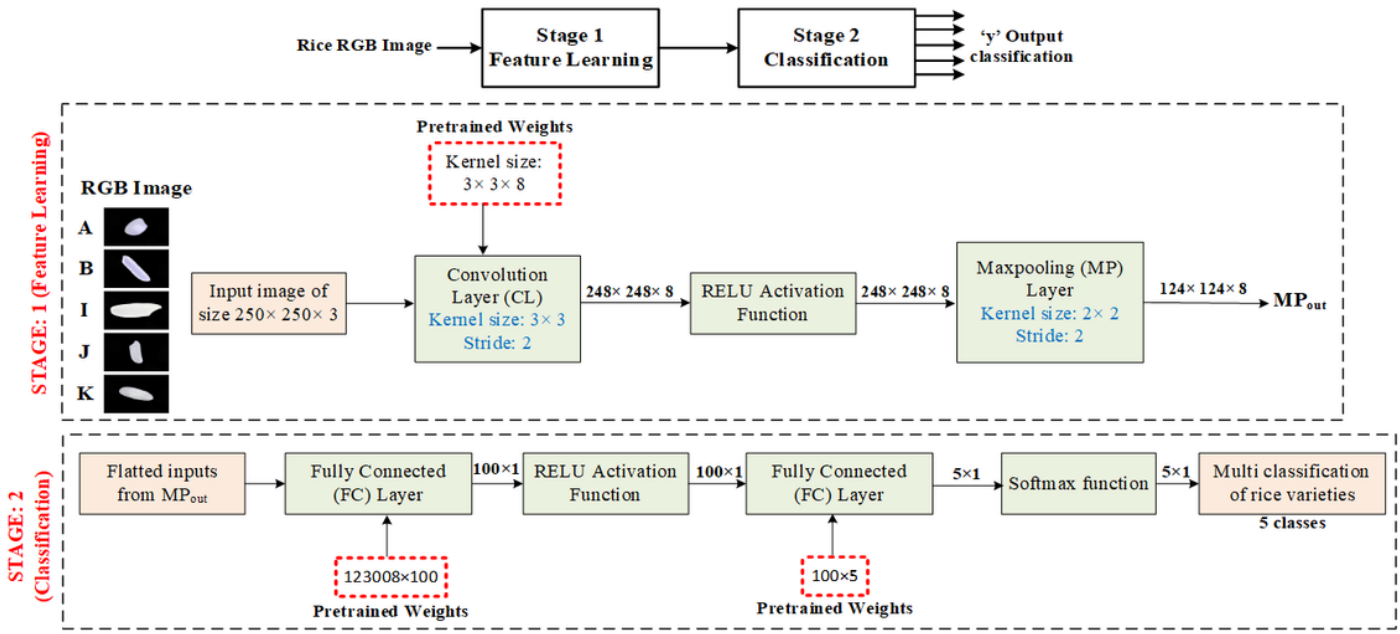

Figure 4

The structure of the proposed Optimised Lightweight CNN (OpLW-CNN) Model.

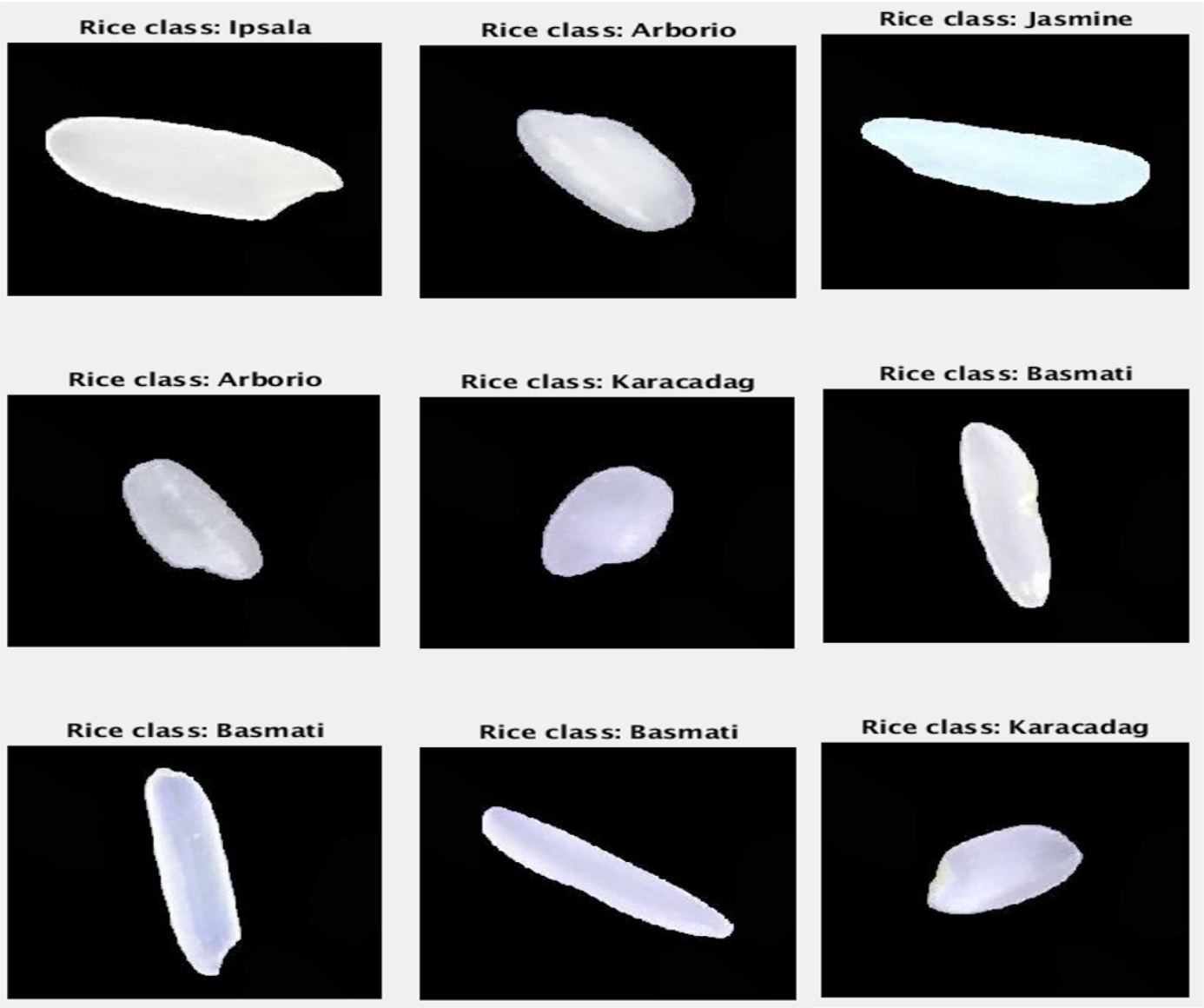

Figure 5

Labeled the rice varieties using the OpLW-CNN model during testing.